

Math 411 Summer 2025**Midterm 1 Review****July 23, 2025****8:30 - 9:30 AM**

Name: _____

Student ID: _____

Instructions:

1. You have 60 minutes to finish the exam.
2. You may only work on your own.
3. Please try to write as legible as you can.
4. If you see a problem, raise your hand and let me know.
5. Write down and sign your honor statement.

Question:	1	2	3	4	Total
Points:	20	20	20	40	100
Score:					

1. Let G be a finite group and R be a commutative ring. Write down the following.
 - (a) (4 points) The statement of the Euclidean algorithm.
 - (b) (4 points) The statement of Bezout's theorem
 - (c) (4 points) An element a in a commutative ring R is a zero-divisor if...
 - (d) (4 points) An subset $S \subseteq R$ is a subring if...
 - (e) (4 points) The statement of Lagrange's theorem for the group G .

2. Find all integer solutions to the following equations.

(a) (10 points) $3x + 8y = 13$

(b) (10 points) $a^2 + b^2 = 7$

Proof. Consider mod 4. Then $a^2 + b^2 \equiv 3 \pmod{4}$. But $a, b \equiv 0, 1, 2, 3 \pmod{4}$ which implies $a^2, b^2 \equiv 0, 1 \pmod{4}$. It follows $a^2 + b^2 \equiv 0$ or $2 \pmod{4}$, which is a contradiction. $\square$

3. Determine if the following is true or false by either giving a proof or a counterexample.

(a) (4 points) Does $3 \mid n^3 - n$ always hold for all integer n ?

Proof. Factor $n^3 - n$ as $n(n^2 - 1) = n(n+1)(n-1)$. Then 3 divides any three consecutive numbers. Alternatively, note $(3, n) = 1$ or 3. If $3 \mid n$, then clearly $3 \mid n^3 - n$. If $(3, n) = 1$, then by Fermat's little theorem, $n^2 \equiv 1 \pmod{3}$ which implies $n^3 \equiv n \pmod{3}$. This is equivalent to $3 \mid n^3 - n$. $\square$

(b) (4 points) Is $\sqrt{6}$ a rational number?

Proof. Suppose $\sqrt{6}$ is rational and write $\sqrt{6} = p/q$ for integers p, q that are relatively prime. Then $6q^2 = p^2$ and this implies $2 \mid p^2$ which implies $2 \mid p$. Then $4 \mid p^2$. Consider $3q^2 = p^2/2$. As $4 \mid p^2$, $p^2/2$ is even. So $2 \mid 3q^2$ which implies $2 \mid q^2$ which means $2 \mid q$. Then p, q are not relatively prime. $\square$

(c) (4 points) If $(a, m) = (b, m) = 1$, then $(ab, m) = 1$ for all nonzero integers a, b , and m .

Proof. Since $(a, m) = 1$, there exists integers x_0, y_0 such that $ax_0 + my_0 = 1$ which implies $ax_0b + my_0b = b$. Since $(b, m) = 1$, there exists integers x_1, y_1 such that $bx_1 + my_1 = 1$. Replace b with $ax_0b + my_0b$. We get

$$abx_0x_1 + m(y_0bx_1 + y_1) = (ax_0b + my_0b)x_1 + my_1 = 1.$$

Bezout's theorem says the least positive integer linear combination of ab and m is (ab, m) . Since 1 is attainable in such way, $(ab, m) = 1$ as 1 is the least positive integer. $\square$

(d) (4 points) A subring S of a domain D is a domain.

(e) (4 points) Is $9^{43} \pmod{23} = 1$?

Proof. Note $(9, 23) = 1$. By Fermat's little theorem, $9^{22} \equiv 1 \pmod{23}$. Then

$$9^{43} = 9^{2 \times 22 - 1} \equiv 9^{-1}.$$

Note $5 \times 9 = 45 \equiv -1 \pmod{23}$. So $22 \times 5 \times 9 \equiv 1 \pmod{23}$ as $22 \equiv -1 \pmod{23}$. Verify $18 \times 9 \equiv 1 \pmod{23}$ and by the uniqueness of inverse in a group $(\mathbb{Z}_{23}^\times)$, we conclude $18 = 9^{-1} \pmod{23}$. So $9^{-1} \equiv 22 \times 5 \equiv 18 \pmod{23}$.

Or one can use Euclidean algorithm to compute $9^{01} \pmod{23}$.

Or note $\mathbb{Z}_{23}^\times$ is a group of order $23 - 1 = 22$ since 23 is a prime. By Lagrange's theorem, the order of 9, denoted by $|9|$, divides 22. So $|9| = 1$ or 11 or 22. But $43 \not\equiv 0 \pmod{11}$ or $\pmod{22}$. This implies $9^{43} \equiv 1 \pmod{23}$ for $9^d \equiv 1 \pmod{23}$ iff 11 or $22 \mid d$. $\square$

4. Consider the commutative ring $\mathbb{Z}_4[i] = \{[a] + [b]i : a, b \in \mathbb{Z}\}$. In particular $i^2 = [-1] = [3]$.

(a) (8 points) What is the order of $\mathbb{Z}_4[i]$ as an abelian group?

Proof. We count directly that $a + bi$ where $a, b \in \mathbb{Z}_4$. Each a, b have 4 choices so there are 16 elements in $\mathbb{Z}_4[i]$. $\square$

(b) (8 points) Give an example of a zero-divisor in $\mathbb{Z}_4[i]$.

Proof. Note $[2] \cdot [2] = [4] = [0]$. $\square$

(c) (8 points) Does $\mathbb{Z}_4[i]$ contains a subring that is a field?

Proof. No. Suppose $S \subseteq \mathbb{Z}_4[i]$ is a subring that is a field. Then $[1] \subseteq S$ which implies $\mathbb{Z}_4 \subseteq S$. But $\mathbb{Z}_4$ is not a domain by part b. $\square$

(d) (8 points) Factorize $[1]$ in two different ways in $\mathbb{Z}_4[i]$ besides $[1] = [1] \cdot [1]$.

Proof. Note

$$[3] \cdot [3] = [9] = [1]$$

$$i \cdot [3]i = [3][-1] = [3][3] = [1]$$

$$([1] - [2]i)^2 = [1] - [2][2]i + [2][2][-1] = [1]$$

$\square$

(e) (8 points) Is $[1] + [2]i$ an unit in $\mathbb{Z}_4[i]$?

Proof. Suppose it is. Then consider $([1] + [2]i)([c] + [d]i) = [1]$. This implies $c - 2d \equiv 1 \pmod{4}$ and $d + 2c \equiv 0 \pmod{4}$. So let $c = 1$ and $d = 2$. So indeed it is a unit. $\square$

(f) (0 points) Is $\mathbb{Z}_4[i]$ an UFD?

Proof. No. Because it is not a domain. $\square$

(g) (0 points) Is $[2]$ prime?

Proof. No. Because $[2] = ([1] + [1]i)([1] + [3]i)$ but clearly $[2] \nmid [1] + i$ nor $[2] \nmid [1] + [3]i$. $\square$

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